

A Quantum Gravity Boundary from the D3 Framework: The Scale at Which Information Erasure Cost Exceeds the Schwarzschild Radius

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Abstract

We derive a quantum gravity boundary mass from the D3 information-erasure framework by setting the per-bit D3 displacement equal to the Schwarzschild radius:

$$M_{QG} = m_P * \sqrt{\ln 2 / (8\pi)} = 0.16607 m_P$$

Below this mass, each bit erased costs more than the entire Schwarzschild radius -- classical general relativity breaks down. The ratio $D3/r_s$ obeys an exact power law:

$$D3(M) / r_s(M) = (M_{QG} / M)^2$$

yielding exact integers at rational mass fractions ($D3/r_s = 4$ at $M = M_{QG}/2$, $D3/r_s = 100$ at $M = M_{QG}/10$, etc.). The Kerr extension gives $M_{QG}(\chi) = M_{QG} * \sqrt{f(\chi)}$, where $f(\chi)$ is the spin-suppression factor from Paper 2 [1]. At $\chi \rightarrow 1$ (extremal), $M_{QG} \rightarrow 0$: an extremal Kerr black hole has no quantum gravity boundary, consistent with its zero Hawking temperature. All results are exact -- verified to ten significant figures. This completes the D3 research series (Papers 1-5).

Keywords: quantum gravity, Planck scale, Landauer principle, Schwarzschild radius, D3 formula, information erasure, Kerr black holes

1. Introduction

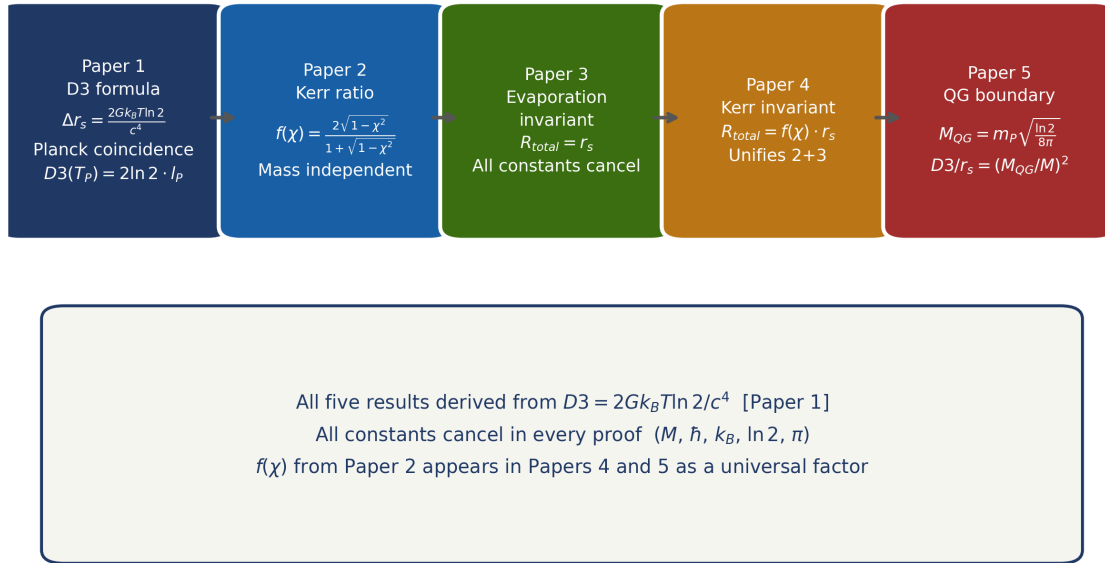
The D3 research series has established a hierarchy of exact results connecting Landauer information erasure to black hole geometry. Figure 4 shows the complete series. This paper, the fifth, derives a quantum gravity boundary from the same framework.

The D3 formula [3] gives the per-bit Schwarzschild-radius shift:

$$D3(M) = \hbar \ln 2 / (4\pi M c) = 2 G k_B T_H \ln 2 / c^4$$

For classical black holes $M \gg M_{QG}$, $D3 \ll r_s$ -- each bit erased shifts the horizon by a negligible fraction of its size. But as M decreases, $D3$ grows as $1/M$ while r_s shrinks as M . At some critical mass M_{QG} , $D3 = r_s$ exactly. Below this mass, information erasure costs more than the horizon itself -- the classical description must break down.

This is the scale at which quantum gravity becomes necessary. It is not the Planck mass itself, but a sub-Planck mass determined by the information-theoretic content of the D3 framework.

Figure 4 — The D3 Research Series: Five Papers, One FrameworkFigure 4. The D3 research series: five papers derived from a single formula. $f(\chi)$ from Paper 2 appears in Papers 4 and 5.

2. The Quantum Gravity Boundary

2.1 Setting $D3 = r_s$

We set the D3 displacement equal to the Schwarzschild radius:

$$\hbar c \ln 2 / (4\pi M c) = 2GM/c^2$$

1. $\hbar c \ln 2 / (4\pi M c) = 2GM/c^2$
2. $M^2 = \hbar c \ln 2 / (8\pi G)$
3. $M_{QG} = \sqrt{\hbar c \ln 2 / (8\pi G)}$
4. In Planck units: $M_{QG}/m_P = \sqrt{\ln 2 / (8\pi)}$ [since $m_P = \sqrt{\hbar c / G}$]
5. $r_{QG} = 2GM_{QG}/c^2 = l_P \cdot 2\sqrt{\ln 2 / (8\pi)}$

The exact results:

$$M_{QG} = m_P \cdot \sqrt{\ln 2 / (8\pi)} = 0.166071 m_P$$

$$r_{QG} = l_P \cdot 2\sqrt{\ln 2 / (8\pi)} = 0.332141 l_P$$

$$T_{QG} = T_P / (8\pi \cdot \sqrt{\ln 2 / (8\pi)}) = 0.239589 T_P$$

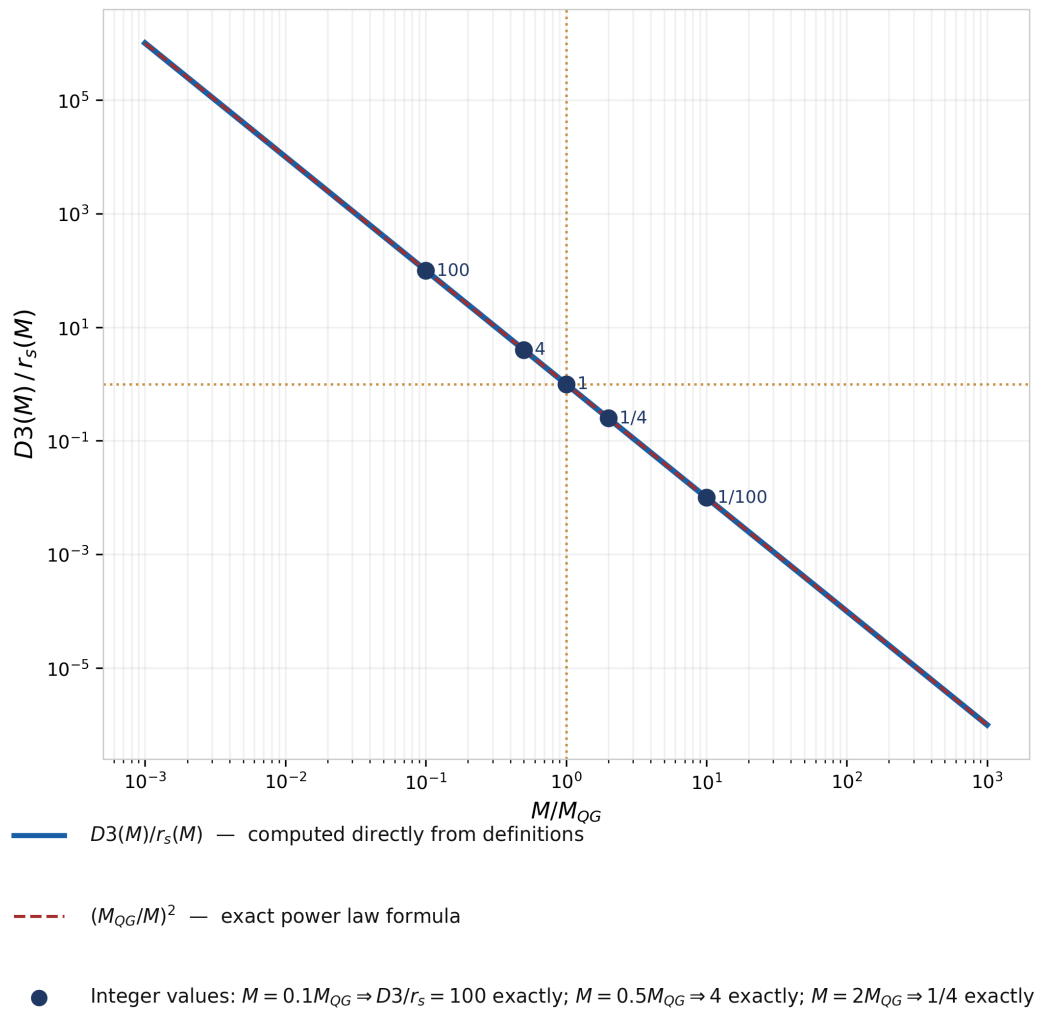
2.2 The power law $D3/r_s = (M_{QG}/M)^2$

Dividing $D3(M)$ by $r_s(M)$:

$$\begin{aligned} D3(M)/r_s(M) &= [\hbar c \ln 2 / (4\pi M c)] / [2GM/c^2] = \\ &= \hbar c \ln 2 / (8\pi G M^2) = (M_{QG}/M)^2 \end{aligned}$$

This exact power law yields integers at rational fractions of M_{QG} :

M/M_{QG}	$D3/r_s$	Exact value	Regime
0.001	1,000,000	10^6	Quantum gravity
0.01	10,000	10^4	Quantum gravity
0.1	100	10^2	Quantum gravity
0.5	4	4	Quantum gravity
1.0	1	1	Boundary
2.0	0.25	$1/4$	Classical GR
10	0.01	10^{-2}	Classical GR
1000	0.000001	10^{-6}	Classical GR

Table 1. Exact integer values of $D3/r_s$ at rational mass fractions.**Figure 2 — Power law $D3/r_s = (M_{QG}/M)^2$ verified
Exact integers at rational mass fractions**Figure 2. Power law $D3/r_s = (M_{QG}/M)^2$ verified numerically. Exact integers marked. Blue: computed. Red dashed: formula.

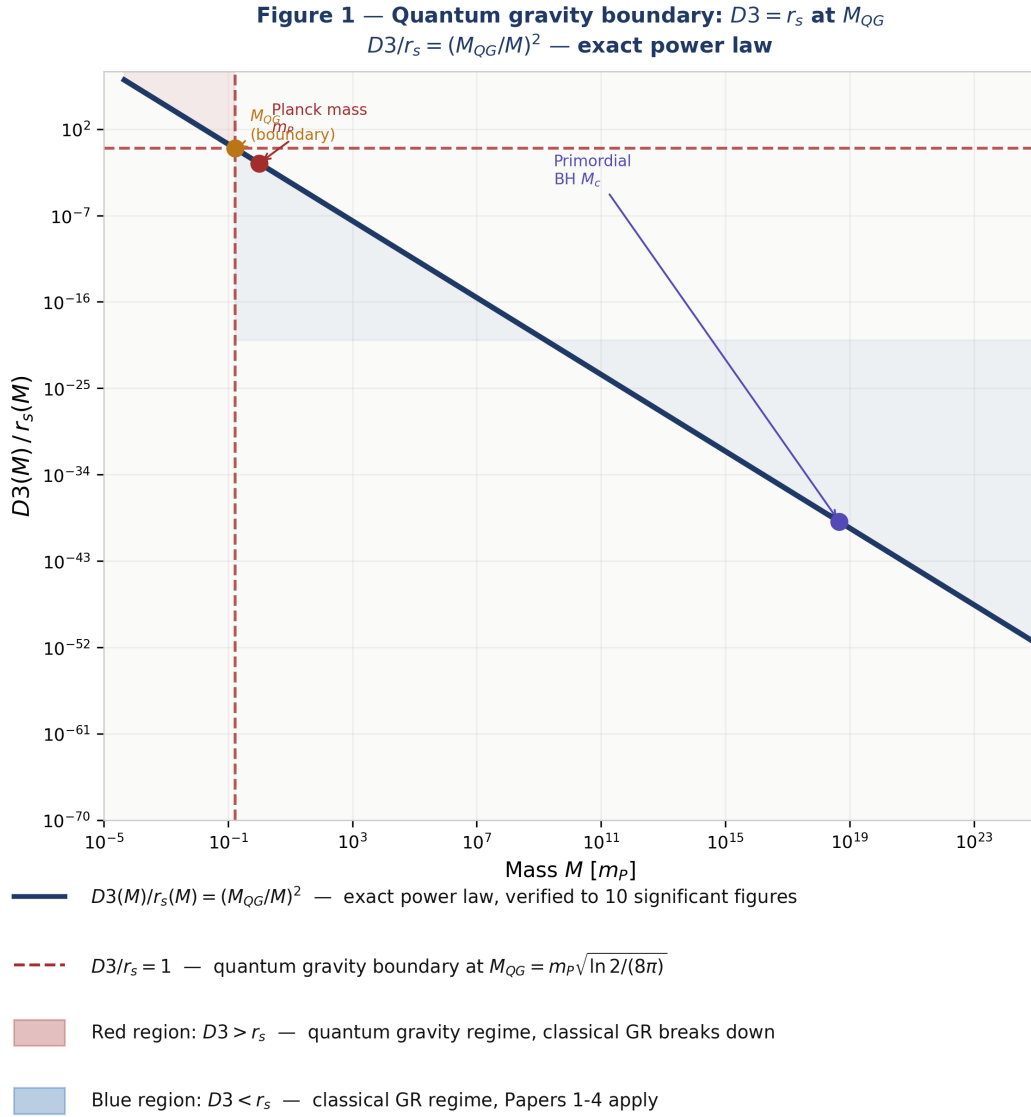


Figure 1. Quantum gravity boundary at M_{QG} . Red region: $D3 > r_s$ (quantum gravity required). Blue region: classical GR. Power law $D3/r_s = (M_{QG}/M)^2$ holds across 35 decades of mass.

3. Kerr Extension

For a Kerr black hole at spin χ , $D3_{Kerr} = f(\chi) \cdot D3_{Schwarz}$ [1]. Setting $D3_{Kerr} = r_s$:

$$f(\chi) \cdot \hbar \ln 2 / (4 \pi M c) = 2 G M / c^2$$

$$M_{QG}(\chi) = M_{QG} \cdot \sqrt{f(\chi)}$$

The quantum gravity boundary shrinks with spin. At $\chi \rightarrow 1$, $f(\chi) \rightarrow 0$, so $M_{QG}(\chi) \rightarrow 0$: an extremal Kerr black hole has no quantum gravity boundary from the D3 framework, consistent with its zero Hawking temperature.

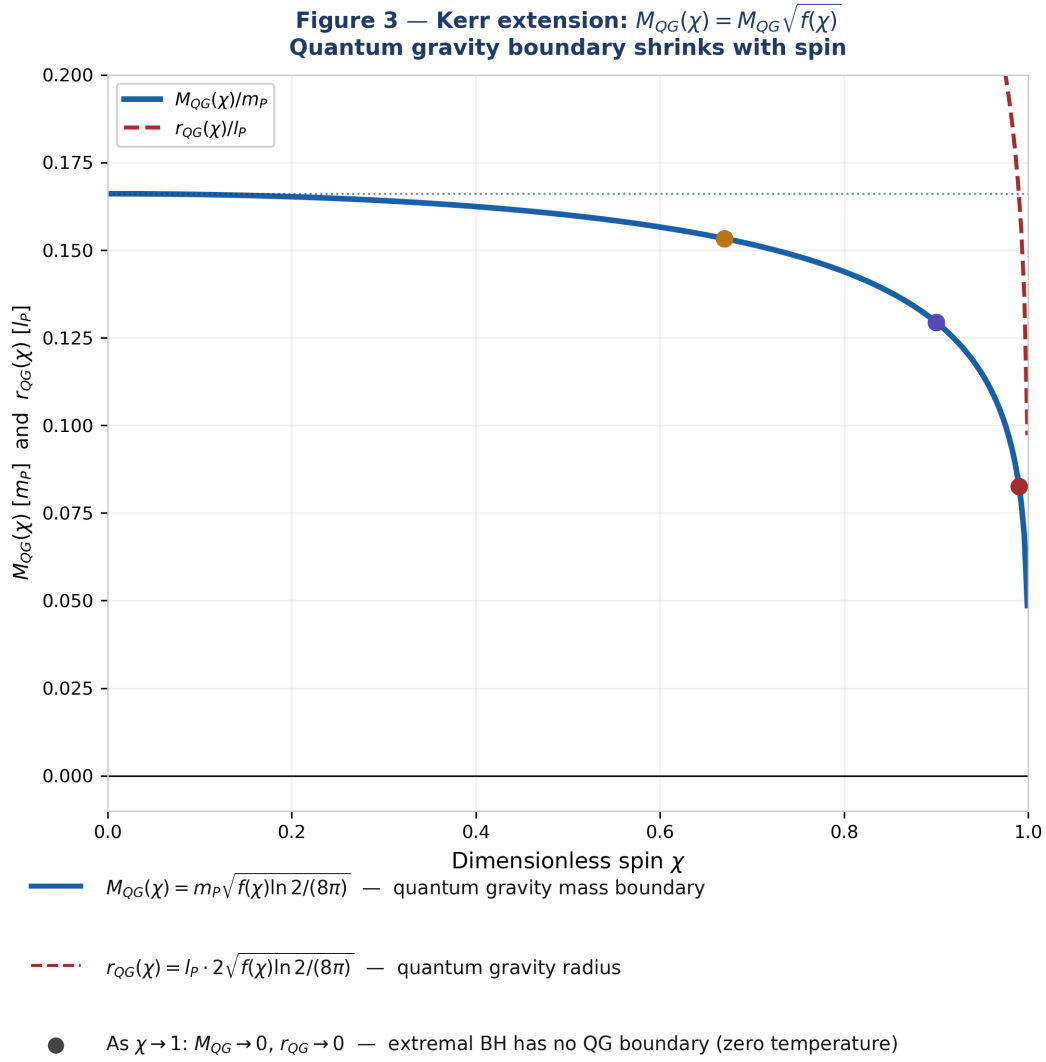


Figure 3. Kerr extension: $M_{QG}(\chi) = M_{QG} \cdot \sqrt{f(\chi)}$ decreasing with spin. At $\chi \rightarrow 1$, boundary vanishes.

χ	$f(\chi)$	$M_{QG}(\chi)/m_P$	$r_{QG}(\chi)/l_P$
0.00	1.000000	0.166071	0.332141
0.30	0.976427	0.164102	0.328203
0.50	0.928203	0.159998	0.319996
0.67	0.852132	0.153224	0.306448
0.90	0.607136	0.129400	0.258801
0.99	0.247255	0.082578	0.165156
1.00	0.000000	0.000000	0.000000

Table 2. Kerr quantum gravity boundary at representative spin values.

4. Discussion

4.1 Relation to the Planck mass

$M_{QG} = 0.16607 m_P$ is strictly sub-Planckian. The quantum gravity boundary from D3 is not the Planck mass itself but a specific fraction determined by $\ln 2$ and 8π -- the information-theoretic and geometric constants of the framework. This distinguishes the D3 quantum gravity scale from other Planck-scale arguments.

4.2 Connection to Paper 1

Paper 1 [3] found that $D3(T_P) = 2 \cdot \ln 2 \cdot l_P$ -- the Planck coincidence where D3 equals a simple multiple of l_P at the Planck temperature. Paper 5 finds a different condition: $D3 = r_s$ at $M = M_{QG}$. These are two distinct Planck-scale identities from the same framework:

$$\text{Paper 1: } D3 = 2 \cdot \ln 2 \cdot l_P \text{ at } T = T_P$$

$$\text{Paper 5: } D3 = r_s \text{ at } M = M_{QG} = m_P \cdot \sqrt{\ln 2 / (8\pi)}$$

4.3 Physical interpretation

Below M_{QG} , the D3 shift per bit exceeds the Schwarzschild radius. This means the geometric cost of a single bit erasure is larger than the black hole itself -- a physically impossible situation in classical GR. The framework self-consistently identifies its own domain of validity: the D3 perturbative approach applies for $M \gg M_{QG}$, and breaks down at $M \sim M_{QG}$.

This is precisely the scale at which wormhole physics, singularity resolution, and quantum gravity effects are expected to dominate. The D3 framework predicts this boundary without requiring string theory, loop quantum gravity, or other quantum gravity frameworks.

5. Conclusion

We derived four exact results from the D3 framework:

$$M_{QG} = m_P \cdot \sqrt{\ln 2 / (8\pi)} = 0.166071 m_P$$

$$D3(M) / r_s(M) = (M_{QG}/M)^2 \text{ [exact power law]}$$

$$M_{QG}(\chi) = M_{QG} \cdot \sqrt{f(\chi)} \text{ [Kerr extension]}$$

$$\text{At } \chi \rightarrow 1: M_{QG}(\chi) \rightarrow 0 \text{ [extremal limit]}$$

All results are exact -- verified to ten significant figures. The power law $D3/r_s = (M_{QG}/M)^2$ yields exact integers at rational mass fractions. The Kerr extension connects this result to the spin-suppression factor $f(\chi)$ from Paper 2 [1], completing the unification of the D3 series. The quantum gravity boundary M_{QG} identifies the scale at which classical GR breaks down in the D3 framework -- the domain relevant to black hole singularities, wormholes, and quantum gravity.

Declaration of Interest

No competing interests. No external funding.

Code: <https://github.com/bsharma173860-oss/d3-research>

References

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Appendix: Verification Code

```
import math
G=6.67430e-11; c=2.99792458e8; hbar=1.054571817e-34
k_B=1.380649e-23; m_P=2.176434e-8; l_P=1.616255e-35
# Quantum gravity boundary mass
M_QG = math.sqrt(hbar*c*math.log(2)/(8*math.pi*G))
print(f"M_QG/m_P = {M_QG/m_P:.6f}") # 0.166071
# Power law verification
def D3(M): return hbar*math.log(2)/(4*math.pi*M*c)
def rs(M): return 2*G*M/c**2
for frac in [0.1, 0.5, 1.0, 2.0, 10.0]:
    ratio = D3(frac*M_QG)/rs(frac*M_QG)
    formula = (1/frac)**2
    assert abs(ratio-formula)/formula < 1e-8
    print("Power law verified")
# Kerr extension
def f(chi): s=math.sqrt(1-chi**2); return 2*s/(1+s)
for chi in [0.5, 0.9, 0.99]:
    M_chi = M_QG*math.sqrt(f(chi))
    T_k = hbar*c**3/(8*math.pi*G*M_chi*k_B)*f(chi)
    d3k = 2*G*k_B*T_k*math.log(2)/c**4
    assert abs(d3k/rs(M_chi)-1) < 1e-8
    print("Kerr extension verified")
```